

TASK 3 – WIRESHARK PACKET ANALYSIS

1. Introduction

Network traffic analysis is an important part of cybersecurity because it allows security professionals to understand how systems communicate with each other. Packet analysis can reveal protocols being used, source and destination systems, connection behavior, application requests, and unusual network activity.

For this task, **Wireshark** was used to capture and analyze network packets in an isolated laboratory environment. The analysis was performed between a Kali Linux machine and a Metasploitable test machine.

The main focus was to analyze:

- HTTP traffic
- DNS traffic
- TCP communication
- TCP three-way handshake
- TCP flags
- Packet structure
- Network-layer and transport-layer information
- Unusual or anomalous traffic
- Wireshark Expert Information

The exercise was performed only within the controlled lab environment.

2. Objective

The objectives of this task were:

1. Capture network traffic using Wireshark.
2. Identify different protocols present in the network traffic.
3. Capture and analyze HTTP traffic.
4. Capture and analyze DNS traffic.
5. Observe TCP communication.
6. Analyze the TCP three-way handshake.
7. Understand TCP flags such as SYN, ACK and FIN.
8. Examine packet structure at different protocol layers.
9. Identify unusual or anomalous network traffic.

10. Use Wireshark's Expert Information feature to identify network events and potential problems.
 11. Document the findings using screenshots and a security analysis report.
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3. Lab Environment

Component	Details
Operating System	Kali Linux
Target System	Metasploitable
Kali IP Address	192.168.56.101
Metasploitable IP Address	192.168.56.102
Network Interface	eth0
Packet Analysis Tool	Wireshark
Network Type	Isolated VirtualBox lab network
Main Protocols Analysed	HTTP, DNS, TCP, ARP, ICMP

The Metasploitable machine was used as the intentionally vulnerable test system, while Kali Linux was used as the analysis machine.

4. Tools Used

4.1 Wireshark

Wireshark is a network protocol analyzer that captures packets travelling through a network interface and allows individual packets to be inspected at different protocol layers.

It was used in this task to:

- Capture packets
- Filter packets
- Identify protocols
- Examine packet headers
- Analyze TCP flags
- Inspect HTTP requests and responses
- Analyze DNS queries and responses

- Review network events using Expert Information

4.2 Kali Linux

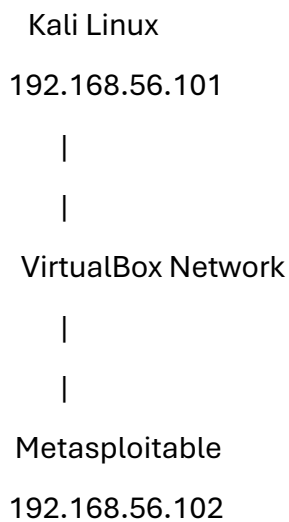
Kali Linux was used as the cybersecurity analysis environment.

4.3 Metasploitable

Metasploitable was used as the intentionally vulnerable laboratory target.

5. Network Topology

The laboratory communication can be represented as:



The traffic captured by Wireshark was generated within this controlled environment.

6. Packet Capture

Wireshark was started on the eth0 interface.

During the capture, different types of traffic appeared, including:

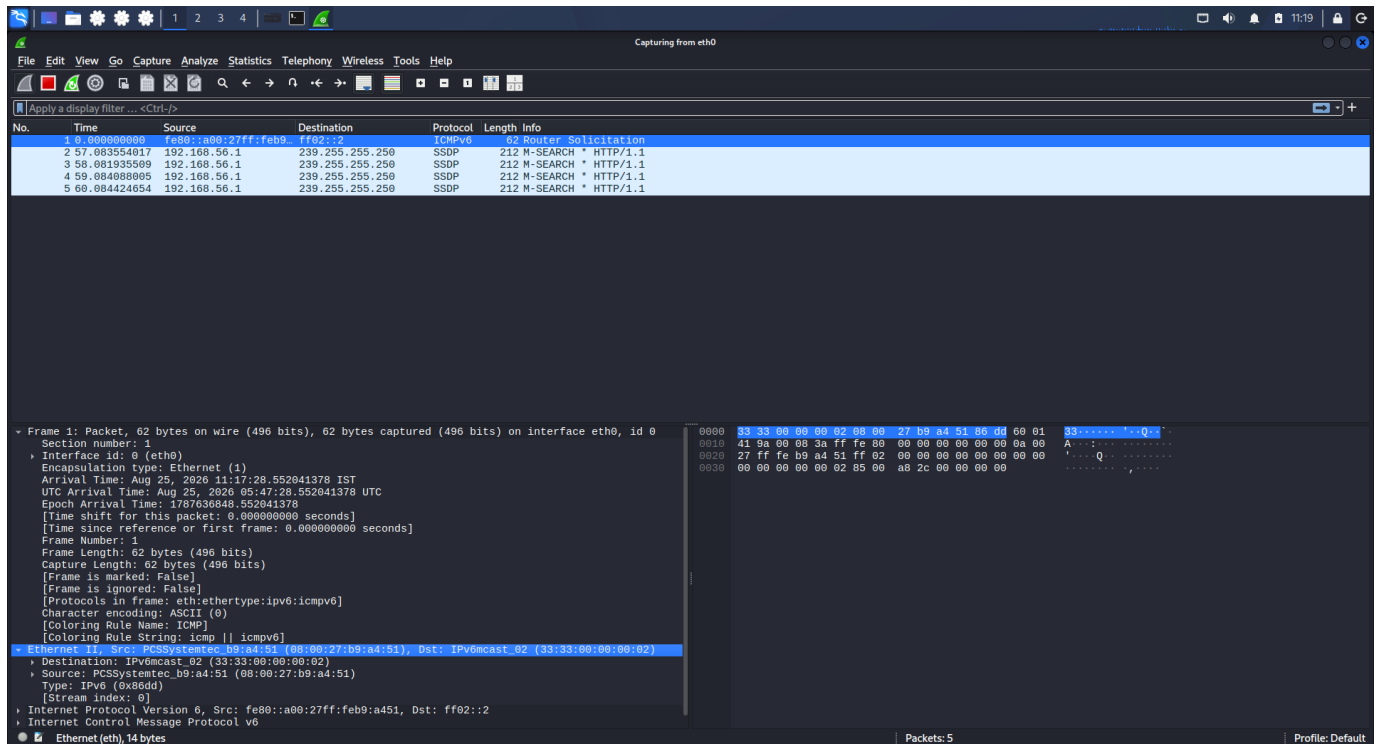
- ARP
- ICMP
- DNS
- TCP
- HTTP
- NBNS
- SSDP
- DHCP

The packet list showed communication between the Kali Linux system and the Metasploitable system.

This confirmed that the network interface was successfully capturing traffic.

Screenshot Evidence

Screenshot:



Observation

The Wireshark interface successfully captured live packets from the eth0 interface. This established the basic packet-capture functionality required for the task.

7. ARP Traffic Analysis

ARP stands for **Address Resolution Protocol**.

It is used to determine the MAC address associated with an IPv4 address on a local network.

During the capture, ARP packets were observed involving:

192.168.56.101

192.168.56.102

Examples of observed traffic included requests similar to:

Who has 192.168.56.102?

Tell 192.168.56.101

This indicates that the Kali machine was attempting to resolve the MAC address associated with the Metasploitable IP address.

Security relevance

ARP is normally required for communication on an IPv4 local network. However, ARP can also be abused in attacks such as ARP spoofing or ARP poisoning.

In this capture, the observed ARP traffic was treated as normal local-network communication.

Screenshot Evidence

The screenshot shows a Wireshark packet capture on interface eth0. The packet list pane displays several packets, including ICMPv6 Router Solicitations, SSDP M-SEARCH requests, and a DHCP Request. The selected packet (No. 10) is a DHCP Request from 192.168.56.100 to 192.168.56.100. The packet details pane shows the Ethernet II header, Internet Protocol Version 4 header, and User Datagram Protocol header. The packet bytes pane shows the raw data in hexadecimal and ASCII.

No.	Time	Source	Destination	Protocol	Length	Info
10	0.000000000	fe80::a00:27ff:feb9::ff02::2	ff02::2	ICMPv6	62	Router Solicitation
2	57.083554017	192.168.56.1	239.255.255.250	SSDP	212	M-SEARCH * HTTP/1.1
3	58.081935509	192.168.56.1	239.255.255.250	SSDP	212	M-SEARCH * HTTP/1.1
4	59.084088005	192.168.56.1	239.255.255.250	SSDP	212	M-SEARCH * HTTP/1.1
5	60.084424654	192.168.56.1	239.255.255.250	SSDP	212	M-SEARCH * HTTP/1.1
6	131.127337502	fe80::a00:27ff:feb9::ff02::2	ff02::2	ICMPv6	62	Router Solicitation
7	157.070453799	PCSSystemtec_b9:a4::	Broadcast	ARP	42	who has 192.168.56.102? Tell 192.168.56.101
8	158.095430922	PCSSystemtec_b9:a4::	Broadcast	ARP	42	who has 192.168.56.102? Tell 192.168.56.101
9	159.119222584	PCSSystemtec_b9:a4::	Broadcast	ARP	42	who has 192.168.56.102? Tell 192.168.56.101
10	160.143668841	PCSSystemtec_b9:a4::	Broadcast	ARP	42	who has 192.168.56.102? Tell 192.168.56.101
11	161.167737294	PCSSystemtec_b9:a4::	Broadcast	ARP	42	who has 192.168.56.102? Tell 192.168.56.101
12	162.192100468	PCSSystemtec_b9:a4::	Broadcast	ARP	42	who has 192.168.56.102? Tell 192.168.56.101
13	172.236256765	PCSSystemtec_b9:a4::	Broadcast	ARP	42	who has 192.168.56.100? Tell 192.168.56.101
14	172.237272414	PCSSystemtec_b7:89::	PCSSystemtec_b9:a4::	ARP	60	192.168.56.100 is at 08:00:27:89:76
15	172.237298455	192.168.56.101	192.168.56.100	DHCP	324	DHCP Request - Transaction ID 0x6ca5eb6b
16	172.243735100	192.168.56.100	192.168.56.101	DHCP	396	DHCP ACK - Transaction ID 0x6ca5eb6b
17	177.141105342	192.168.56.1	239.255.255.250	SSDP	212	M-SEARCH * HTTP/1.1
18	178.142628828	192.168.56.1	239.255.255.250	SSDP	212	M-SEARCH * HTTP/1.1
19	179.144292337	192.168.56.1	239.255.255.250	SSDP	212	M-SEARCH * HTTP/1.1
20	180.144092985	192.168.56.1	239.255.255.250	SSDP	212	M-SEARCH * HTTP/1.1

Frame 10: Packet, 596 bytes on wire (4720 bits), 596 bytes captured (4720 bits) on interface eth0, id 0
Ethernet II, Src: PCSSystemtec_b7:89:76 (08:00:27:89:76), Dst: PCSSystemtec_b9:a4:51 (08:00:27:b9:a4:51)
Destination: PCSSystemtec_b9:a4:51 (08:00:27:b9:a4:51)
.....0..... = LG bit: Globally unique address (factory default)
.....0..... = LG bit: Individual address (unicast)
Source: PCSSystemtec_b7:89:76 (08:00:27:89:76)
Type: IPv4 (0x0800)
[Stream index: 3]
Internet Protocol Version 4, Src: 192.168.56.100, Dst: 192.168.56.101
User Datagram Protocol, Src Port: 67, Dst Port: 68
Dynamic Host Configuration Protocol (ACK)

8. DNS Traffic Analysis

DNS stands for **Domain Name System**.

DNS translates domain or host names into IP addresses.

```
(sourabh@kali)-[~]
$ ping -c 4 192.168.56.102
PING 192.168.56.102 (192.168.56.102) 56(84) bytes of data.
64 bytes from 192.168.56.102: icmp_seq=1 ttl=64 time=2.93 ms
64 bytes from 192.168.56.102: icmp_seq=2 ttl=64 time=2.17 ms
64 bytes from 192.168.56.102: icmp_seq=3 ttl=64 time=1.75 ms
64 bytes from 192.168.56.102: icmp_seq=4 ttl=64 time=2.02 ms

— 192.168.56.102 ping statistics —
4 packets transmitted, 4 received, 0% packet loss, time 3014ms
rtt min/avg/max/mdev = 1.750/2.215/2.927/0.437 ms
```

Wireshark interface showing a packet capture on eth0. The packet list pane displays a list of captured packets, including DNS queries and responses. The packet details pane shows the structure of the selected packet (Frame 90), including Ethernet II, Internet Protocol Version 4, and Internet Control Message Protocol (ICMP) fields. The packet bytes pane displays the raw data of the selected packet.

No.	Time	Source	Destination	Protocol	Length	Info
73	55.063668891	192.168.56.102	192.168.56.255	NBNS	110	Registration NB WORKGROUP<00>
74	55.063989458	192.168.56.102	192.168.56.255	NBNS	110	Registration NB WORKGROUP<1e>
75	55.134139833	192.168.56.1	224.0.0.251	MDNS	240	Standard query response 0x0000 A, cache flush 0.0.0.0 A, cache flush 192.168.1.2 AAAA, cache flush ::
76	55.134140463	Fe80::2bb7:74c0:5ce...	ff02::fb	MDNS	260	Standard query response 0x0000 A, cache flush 0.0.0.0 A, cache flush 192.168.1.2 AAAA, cache flush ::
77	55.061699413	192.168.56.102	192.168.56.255	NBNS	110	Registration NB METASPLOITABLE<20>
78	55.062277497	192.168.56.102	192.168.56.255	NBNS	110	Registration NB METASPLOITABLE<03>
79	55.062522428	192.168.56.102	192.168.56.255	NBNS	110	Registration NB METASPLOITABLE<00>
80	55.063024444	192.168.56.102	192.168.56.255	NBNS	110	Registration NB WORKGROUP<00>
81	55.063024570	192.168.56.102	192.168.56.255	NBNS	110	Registration NB WORKGROUP<1e>
82	55.074556930	Fe80::a90:27ff:fe0e...	ff02::2	ICMPv6	70	Router Solicitation from 08:00:27:0e:5f:8c
83	605.678180948	PCSSystemtec_b9:a4:...	Broadcast	ARP	42	Who has 192.168.56.102? Tell 192.168.56.101
84	605.680202392	PCSSystemtec_b9:a4:...	ARP	60	192.168.56.102 is at 08:00:27:0e:5f:8c	
85	605.680218543	192.168.56.101	192.168.56.102	ICMP	98	Echo (ping) request id=0xc94a, seq=1/256, ttl=64 (reply in 86)
86	605.680962232	192.168.56.102	192.168.56.101	ICMP	98	Echo (ping) reply id=0xc94a, seq=1/256, ttl=64 (request in 85)
87	605.680355301	192.168.56.101	192.168.56.102	ICMP	98	Echo (ping) request id=0xc94a, seq=2/512, ttl=64 (reply in 88)
88	605.680490656	192.168.56.102	192.168.56.101	ICMP	98	Echo (ping) reply id=0xc94a, seq=2/512, ttl=64 (request in 87)
89	607.691047209	192.168.56.101	192.168.56.102	ICMP	98	Echo (ping) request id=0xc94a, seq=3/768, ttl=64 (reply in 90)
90	607.692707448	192.168.56.102	192.168.56.101	ICMP	98	Echo (ping) reply id=0xc94a, seq=3/768, ttl=64 (request in 89)
91	608.69342241	192.168.56.101	192.168.56.102	ICMP	98	Echo (ping) request id=0xc94a, seq=4/1024, ttl=64 (reply in 92)
92	608.69522086	192.168.56.102	192.168.56.101	ICMP	98	Echo (ping) reply id=0xc94a, seq=4/1024, ttl=64 (request in 91)
93	610.679964183	PCSSystemtec_0e:5f:...	PCSSystemtec_b9:a4:...	ARP	60	Who has 192.168.56.101? Tell 192.168.56.102
94	610.679965600	PCSSystemtec_b9:a4:...	PCSSystemtec_0e:5f:...	ARP	42	192.168.56.101 is at 08:00:27:b9:a4:51
95	627.106546493	192.168.56.102	192.168.56.255	BROWSER	286	Host Announcement METASPLOITABLE, Workstation, Server, Print Queue Server, Xenix Server, NT Workstation, NT Server, Potential Bro...

Frame 90: Packet, 98 bytes on wire (784 bits), 98 bytes captured (784 bits) on interface eth0, id 0
Ethernet II, Src: PCSSystemtec_0e:5f:8c (08:00:27:0e:5f:8c), Dst: PCSSystemtec_b9:a4:51 (08:00:27:b9:a4:51)
Destination: PCSSystemtec_b9:a4:51 (08:00:27:b9:a4:51)
.....0..... = LG bit: Globally unique address (factory default)
.....0..... = IG bit: Individual address (unicast)
- Source: PCSSystemtec_0e:5f:8c (08:00:27:0e:5f:8c)
.....0..... = LG bit: Globally unique address (factory default)
.....0..... = IG bit: Individual address (unicast)
Type: IPv4 (0x0800)
[Stream index: 11]
Internet Protocol Version 4, Src: 192.168.56.102, Dst: 192.168.56.101
Internet Control Message Protocol
Type: Echo (ping) reply (0)
Code: 0
Checksum: 0xa490 [correct]
[Checksum Status: Good]
Identifier (BE): 51530 (0xc94a)
Identifier (LE): 19145 (0x4ac9)
Sequence Number (BE): 3 (0x0003)
Sequence Number (LE): 768 (0x0300)
[Request: frame 89]
[Response time: 1.720 ms]
ICMP Data: d02e8d6a0000000072b50300000000101112131415161718191a1b1c1d1e1f202122232425262728292a2b2c

A DNS query was captured between:

Source: 192.168.56.101

Destination: 192.168.56.102

Destination Port: 53

Protocol: UDP

The query requested:

metasploitable.localdomain

Important packet information

The captured packet showed:

User Datagram Protocol

Source Port: 59845

Destination Port: 53

The DNS section showed:

Domain Name System (query)

Transaction ID: 0x4baf

Flags: 0x0120 Standard query

Questions: 1

The packet was therefore a standard DNS query.

DNS Response

The corresponding response from the Metasploitable system indicated:

Server failure

This is an important observation because the requested hostname could not be successfully resolved by the DNS service.

Security Analysis

A DNS Server Failure does not automatically mean that the system has been compromised.

Possible causes include:

- Incorrect DNS configuration
- DNS service problems
- Missing DNS records
- Incorrect local-domain configuration
- Temporary DNS service failure

In a real production environment, repeated DNS failures should be investigated.

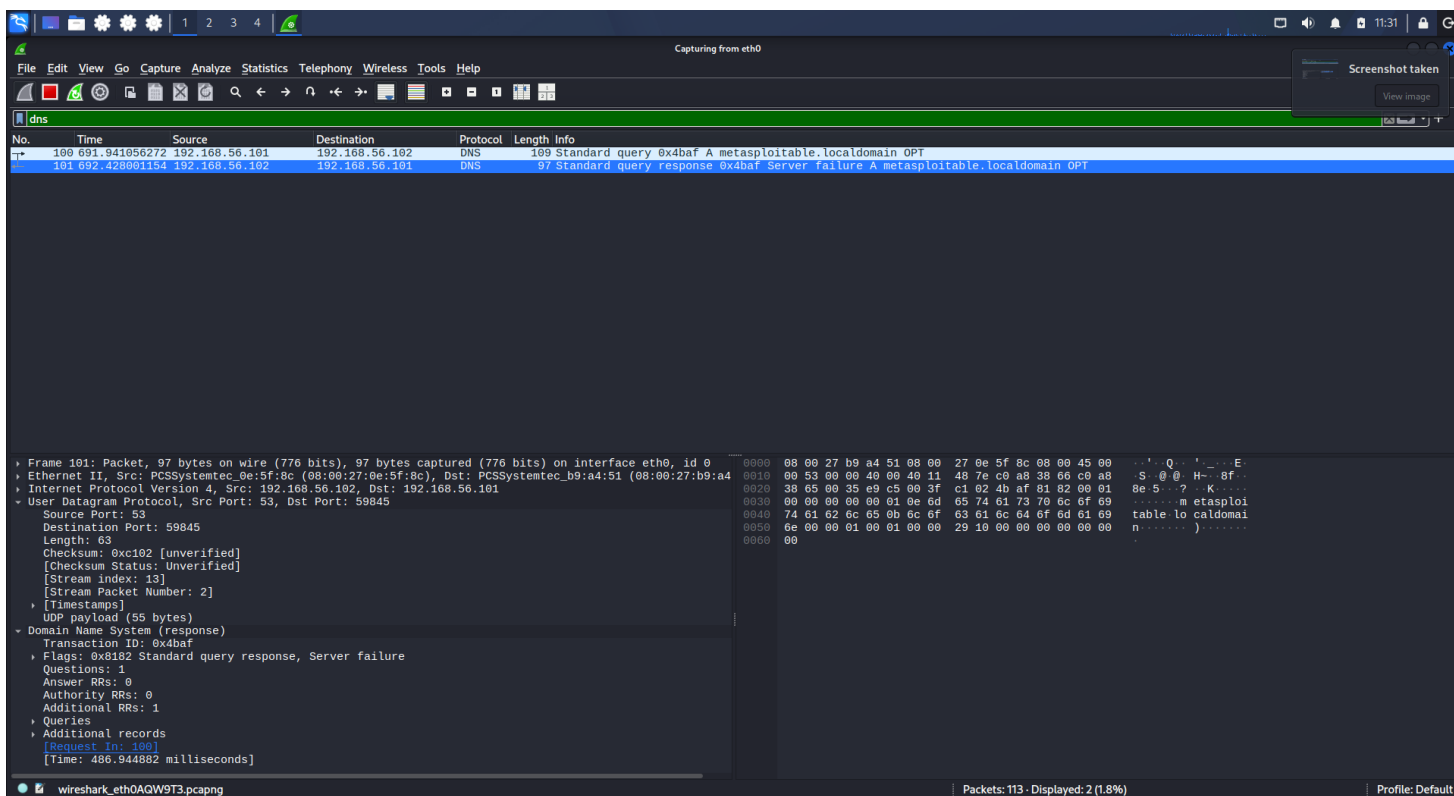
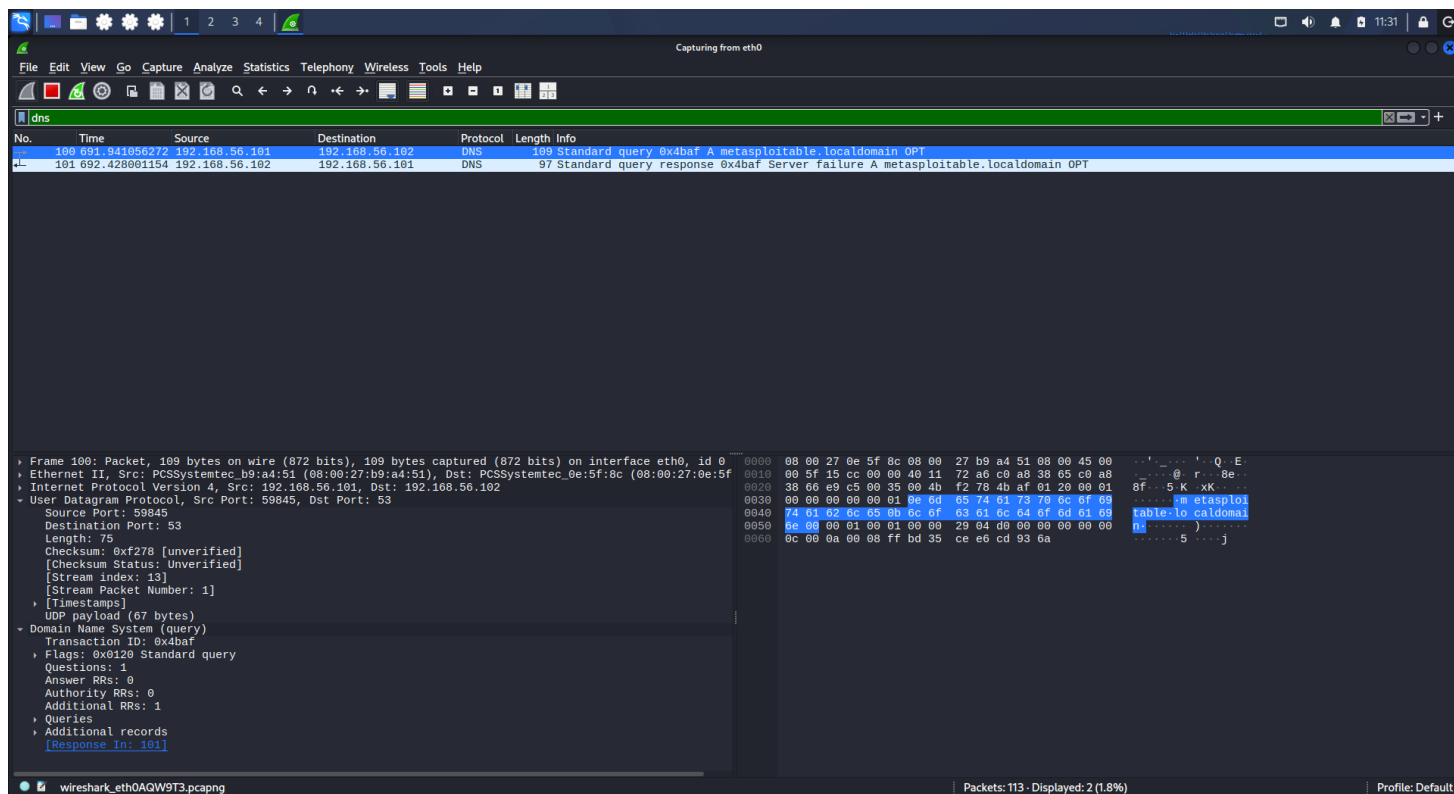
Screenshot Evidence

```
(sourabh@kali)-[~]
$ dig @192.168.56.102 metasploitable.localdomain

; <<>> DiG 9.20.26-1-Debian <<>> @192.168.56.102 metasploitable.localdomain
; (1 server found)
;; global options: +cmd
;; Got answer:
;; ->HEADER<- opcode: QUERY, status: SERVFAIL, id: 19375
;; flags: qr rd ra; QUERY: 1, ANSWER: 0, AUTHORITY: 0, ADDITIONAL: 1

;; OPT PSEUDOSECTION:
; EDNS: version: 0, flags:; udp: 4096
;; QUESTION SECTION:
;metasploitable.localdomain.      IN      A

;; Query time: 488 msec
;; SERVER: 192.168.56.102#53(192.168.56.102) (UDP)
;; WHEN: Tue Aug 25 11:29:00 IST 2026
;; MSG SIZE rcvd: 55
```



9. TCP Traffic Analysis

TCP stands for **Transmission Control Protocol**.

TCP provides reliable, connection-oriented communication between two systems.

During the Wireshark capture, TCP communication was observed between:

192.168.56.101

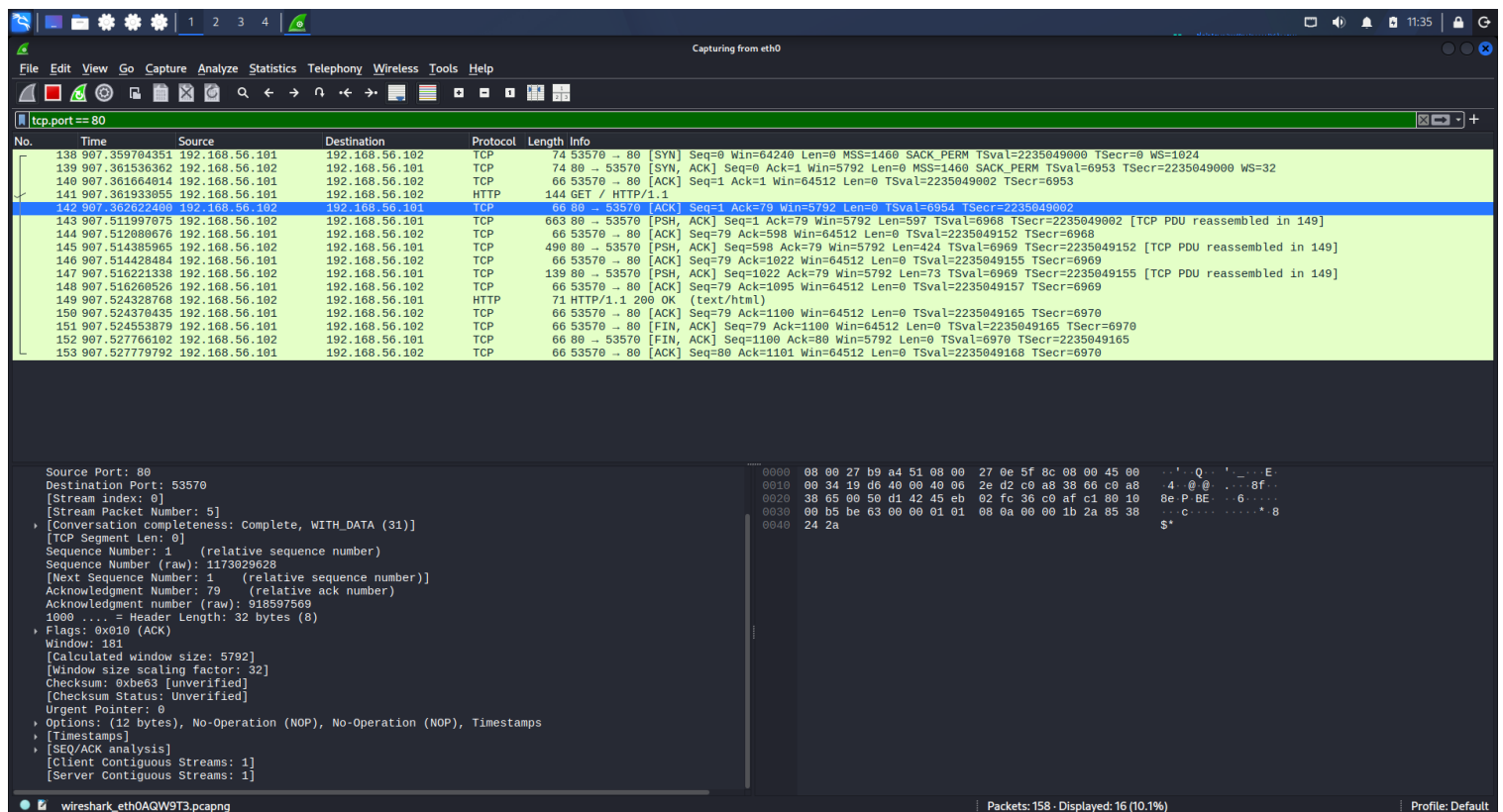
and

192.168.56.102

HTTP communication was observed over TCP port:

80

The captured packets showed TCP connection establishment followed by application-layer HTTP communication.



10. TCP Three-Way Handshake

One of the most important parts of the analysis was observing the TCP three-way handshake.

The handshake consists of three main packets:

Step 1 – SYN

The client sends a SYN packet to request a TCP connection.

192.168.56.101 → 192.168.56.102

[SYN]

The SYN flag indicates that the client wants to establish a connection

The image shows a Wireshark packet capture of a network traffic. The top pane displays a list of captured packets. Packet 138 is highlighted, showing a SYN packet from 192.168.56.101 to 192.168.56.102 on port 80. The packet details pane on the left shows the structure of the packet, including the Ethernet II header, Internet Protocol Version 4 header, and Transmission Control Protocol header. The packet bytes pane on the right shows the raw data of the packet in hexadecimal and ASCII.

No.	Time	Source	Destination	Protocol	Length	Info
138	907.359704351	192.168.56.101	192.168.56.102	TCP	74	53570 → 80 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM TSval=2235049080 TSecr=0 WS=1024

Packet 138 details:

- Ethernet II, Src: PCSSystemtec_b9:a4:51 (08:00:27:b9:a4:51), Dst: PCSSystemtec_0e:5f:8c (08:00:27:0e:5f:8c)
- Internet Protocol Version 4, Src: 192.168.56.101, Dst: 192.168.56.102
- Transmission Control Protocol, Src Port: 53570, Dst Port: 80, Seq: 0, Len: 0
 - Source Port: 53570
 - Destination Port: 80
 - [Stream Index: 0]
 - [Stream Packet Number: 1]
 - [Conversation completeness: Complete, WITH_DATA (31)]
 - [TCP Segment Len: 0]
 - Sequence Number: 0 (relative sequence number)
 - Sequence Number (raw): 918597490
 - [Next Sequence Number: 1 (relative sequence number)]
 - Acknowledgment Number: 0
 - Acknowledgment number (raw): 0
 - 1010 ... = Header Length: 40 bytes (10)
 - Flags: 0x002 (SYN)
 - Window: 64240
 - [Calculated window size: 64240]
 - Checksum: 0xf24a [unverified]
 - [Checksum Status: Unverified]
 - Urgent Pointer: 0
 - Options: (20 bytes), Maximum segment size, SACK permitted, Timestamps, No-Operation (NOP), Window sc
 - [Timestamps]

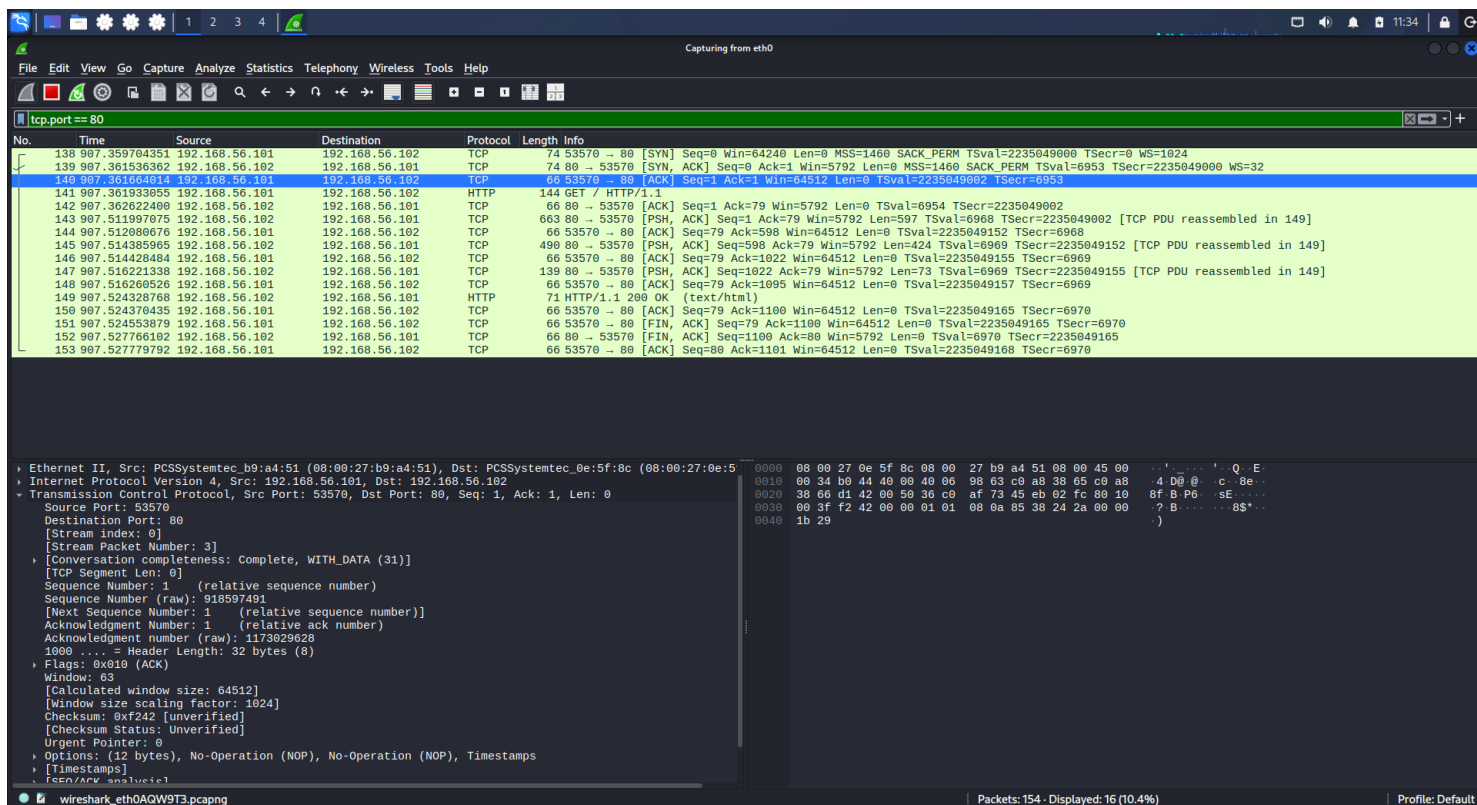
Step 2 – SYN/ACK

The server responds:

192.168.56.102 → 192.168.56.101

[SYN, ACK]

The SYN indicates that the server is also establishing the connection, while ACK acknowledges the client's SYN.



Security relevance

Understanding the TCP handshake is important because abnormal SYN patterns can sometimes indicate scanning or denial-of-service activity.

In this capture, the observed handshake was consistent with normal TCP connection establishment.

11. TCP Flags Analysis

TCP flags provide information about the state and purpose of TCP packets.

The capture showed several important flags.

SYN

Used to establish a TCP connection.

[SYN]

ACK

Used to acknowledge received data or connection requests.

[ACK]

SYN + ACK

Used by the server to acknowledge the client's SYN while also establishing its side of the connection.

[SYN, ACK]

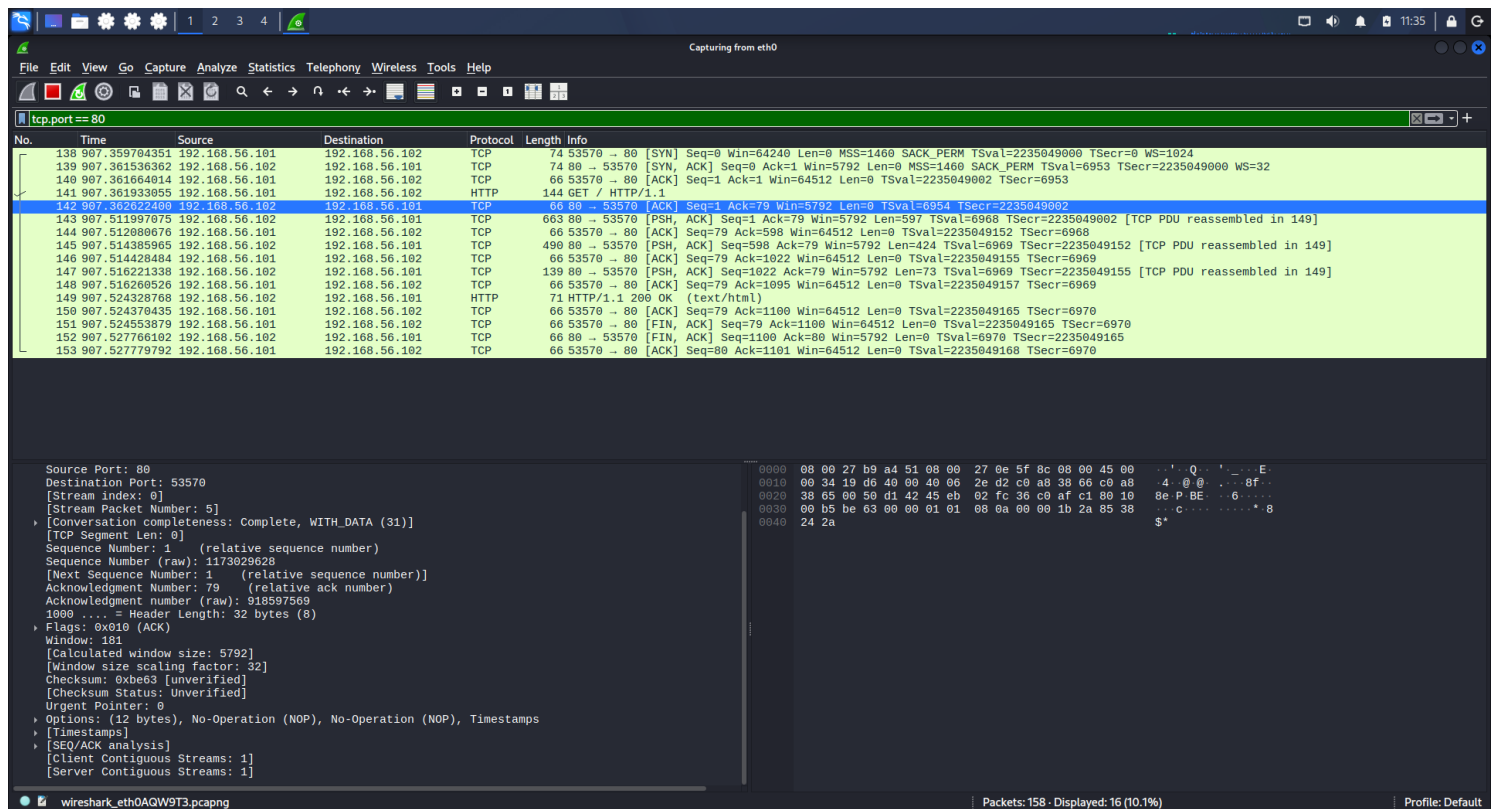
FIN

Used to indicate that a system wants to terminate a TCP connection.

The Wireshark capture showed FIN-related packets during connection termination.

FIN + ACK

Indicates that a connection is being closed and the FIN is being acknowledged.



12. HTTP Traffic Analysis

HTTP stands for **H**ypertext **T**ransfer **P**rotocol.

The captured HTTP traffic showed communication between:

192.168.56.101 → 192.168.56.102

The captured request was:

GET / HTTP/1.1

The request contained:

Host: 192.168.56.102

User-Agent: curl/8.21.0

```
(sourabh@kali)-[~]
$ curl http://192.168.56.102
<html><head><title>Metasploitable2 - Linux</title></head><body>
<pre>

Metasploitable2

Warning: Never expose this VM to an untrusted network!

Contact: msfdev[at]metasploit.com

Login with msfadmin/msfadmin to get started

</pre>
<ul>
<li><a href="/twiki/">TWiki</a></li>
<li><a href="/phpMyAdmin/">phpMyAdmin</a></li>
<li><a href="/mutillidae/">Mutillidae</a></li>
<li><a href="/dvwa/">DVWA</a></li>
<li><a href="/dav/">WebDAV</a></li>
</ul>
</body>
</html>

(sourabh@kali)-[~]
$ █
```

Accept: */*

The Metasploitable system responded with:

HTTP/1.1 200 OK

Server information

The response exposed:

Server: Apache/2.2.8 (Ubuntu) DAV/2

X-Powered-By: PHP/5.2.4-2ubuntu5.10

Content-Type: text/html

The HTTP response also contained HTML content.

Security significance

The capture demonstrates that HTTP traffic can expose information about the server and application.

The server response revealed:

- Web server technology
- Server version

- PHP version
- HTTP response status
- Content type

Additionally, HTTP is not encrypted.

Therefore, information transmitted over plain HTTP can potentially be inspected by someone who can capture the traffic.

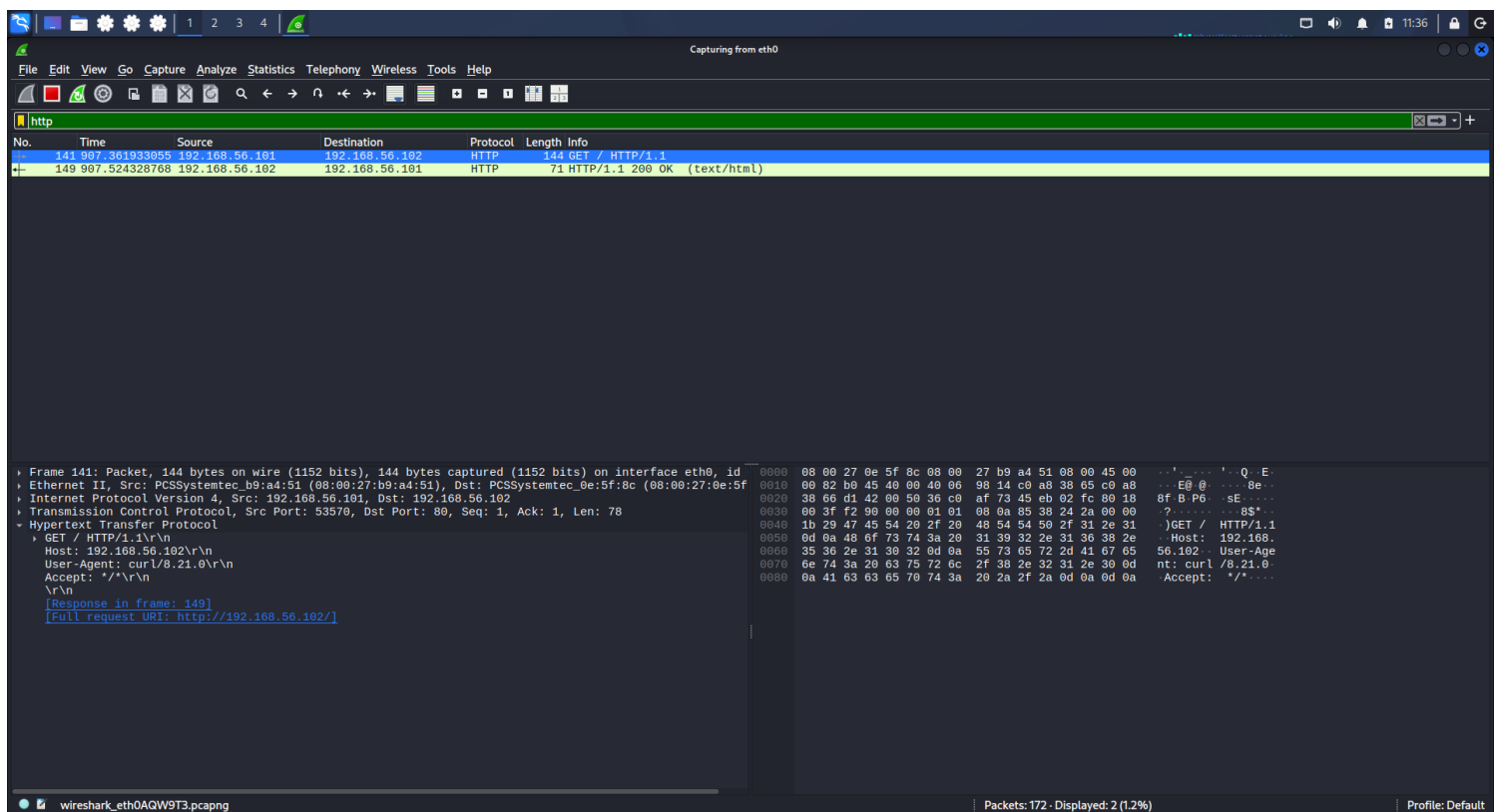
Recommendation

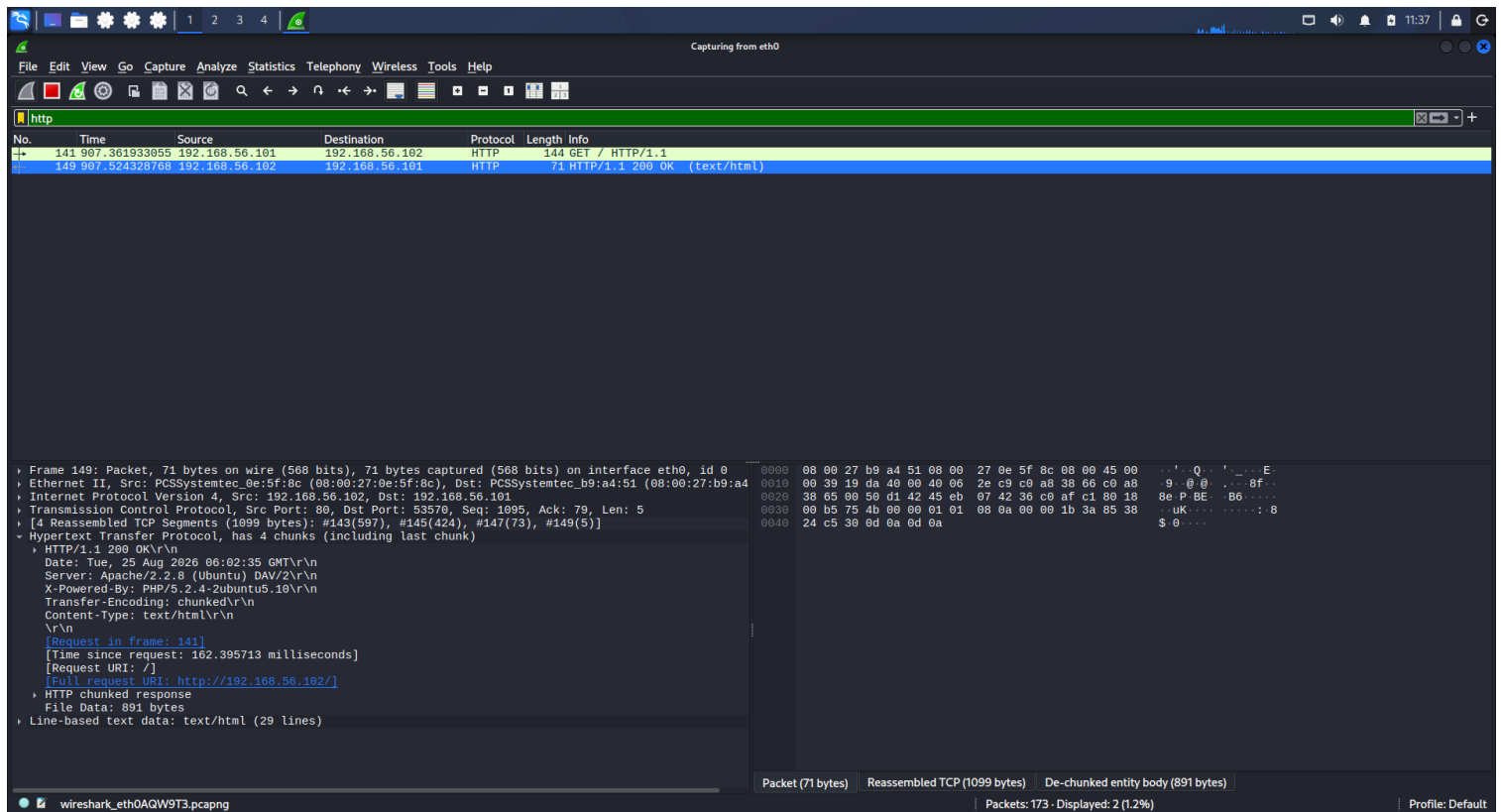
Production systems should use:

HTTPS

instead of unencrypted HTTP wherever sensitive information is transmitted.

Screenshot Evidence





13. Packet Structure Analysis

Wireshark allows packets to be examined layer by layer.

For example, the captured HTTP packet contained:

Frame

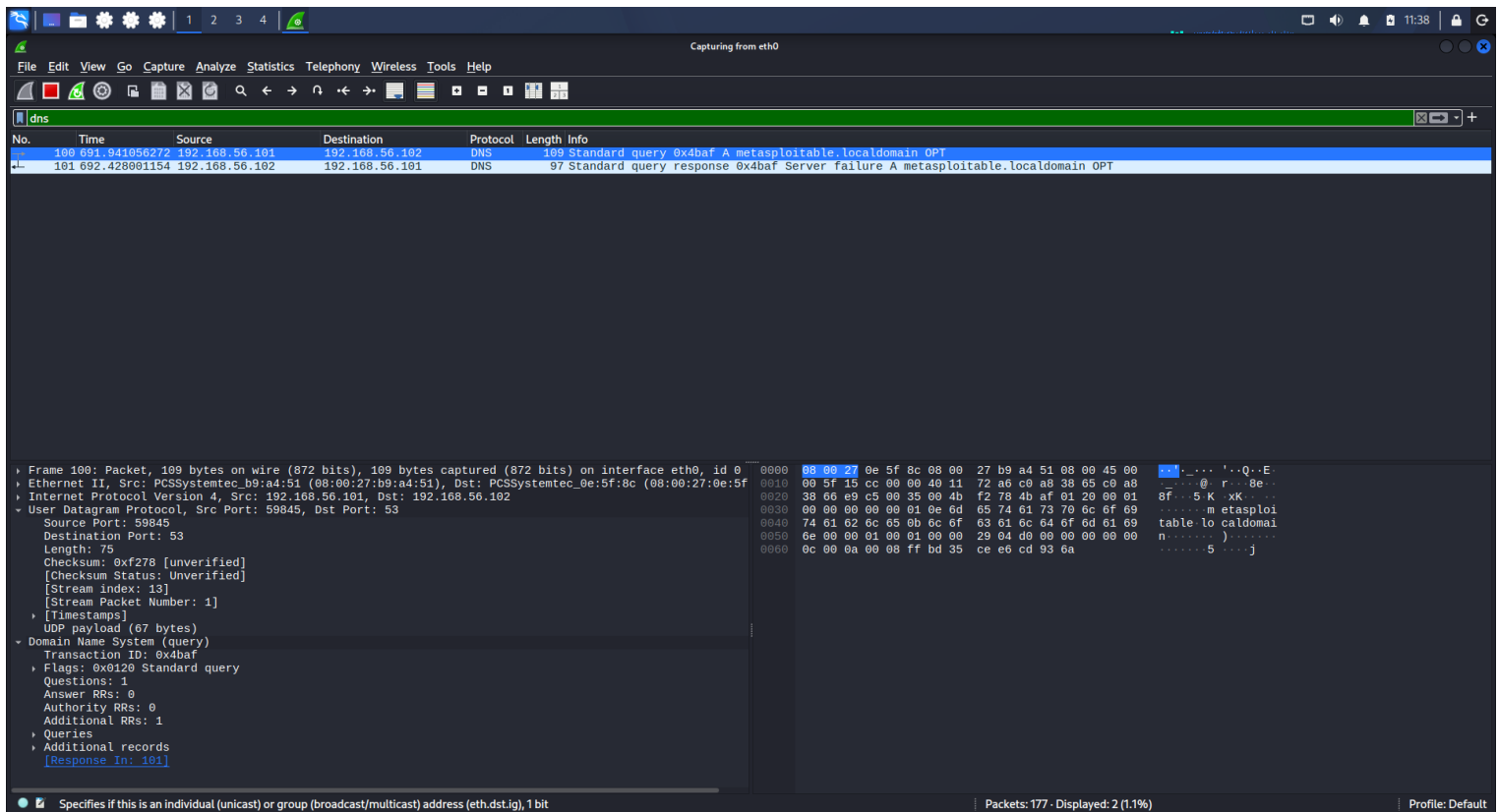
- └─ Ethernet II
 - └─ Internet Protocol Version 4
 - └─ Transmission Control Protocol
 - └─ Hypertext Transfer Protocol

The DNS packet contained:

Frame

- └─ Ethernet II
 - └─ IPv4
 - └─ UDP
 - └─ DNS

This demonstrates how different network protocols are encapsulated within each other.



14. Ethernet Layer

The Ethernet section provides information such as:

- Source MAC address
- Destination MAC address
- Ethernet type

This is the data-link layer information used for communication on the local network.

15. IPv4 Layer

The IPv4 section provides:

- Source IP address
- Destination IP address
- Protocol information
- TTL
- Other IP header information

For example:

Source: 192.168.56.101

Destination: 192.168.56.102

16. Transport Layer

The transport protocol depended on the traffic being analyzed.

For HTTP:

TCP

For DNS:

UDP

The TCP packets provided information about:

- Source port
- Destination port
- Sequence number
- Acknowledgment number
- TCP flags
- Window size

The DNS packet provided:

- Source port
 - Destination port 53
 - UDP payload
 - DNS transaction information
-

17. Unusual / Anomalous Traffic

The capture was reviewed for unusual or malformed traffic.

Wireshark's Expert Information feature was used for this purpose.

The Expert Information window showed:

- TCP connection establishment request
- TCP connection establishment acknowledgement
- TCP connection finish
- Connection closing

- Time To Live information

Importantly, the reviewed Expert Information window did **not show Error or Warning entries**.

Therefore, there is no evidence from this capture that a malformed packet was detected.

Observed anomaly

The strongest unusual event in the capture was the DNS response:

Server failure

for:

metasploitable.localdomain

This was documented as an anomalous DNS event.

It should not be described as a confirmed attack or compromise.

18. Wireshark Expert Information

Wireshark's Expert Information feature provides additional information about packets and protocol behavior.

The captured results showed:

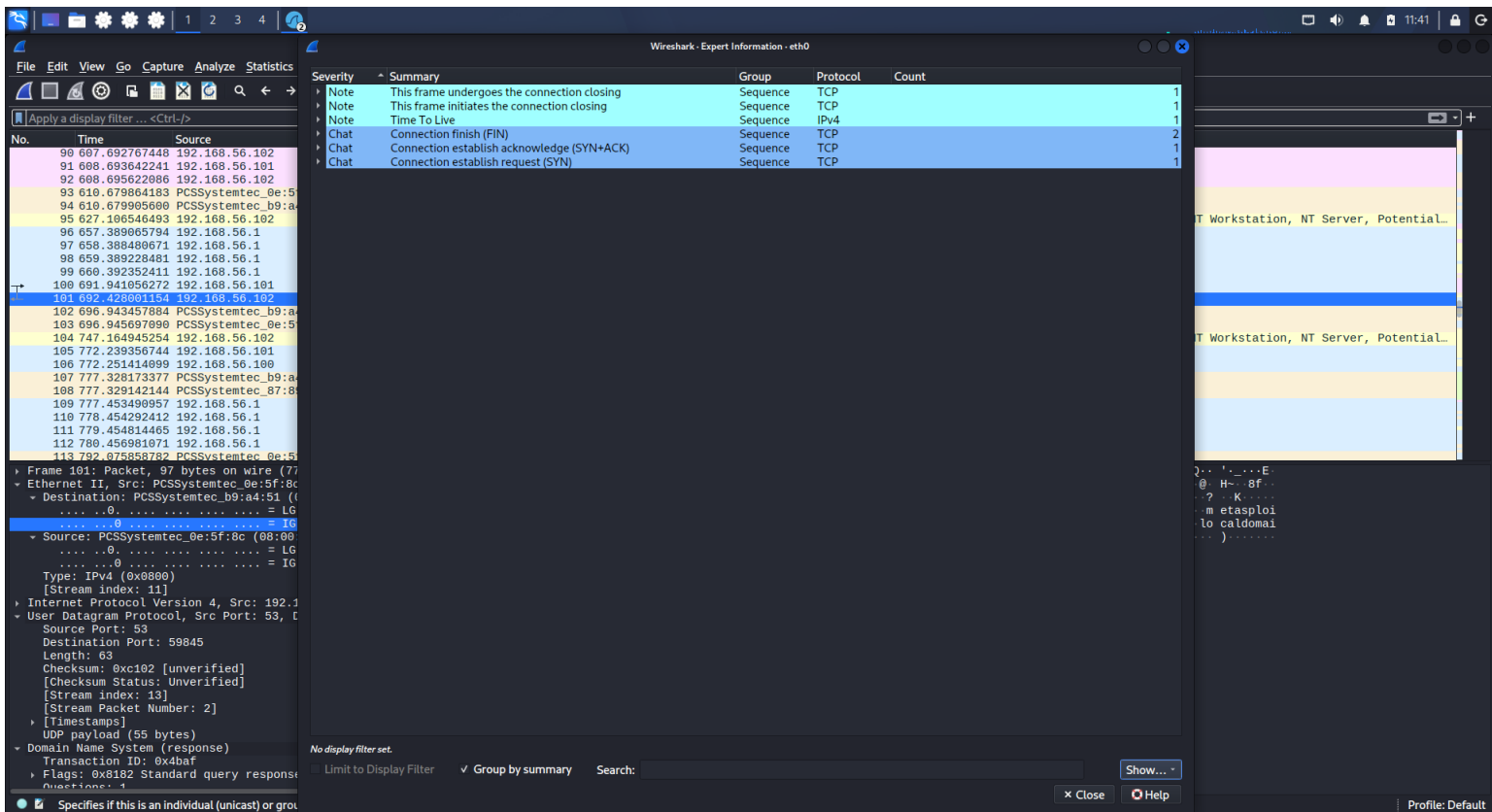
Observation	Protocol Count	
Frame undergoes connection closing	TCP	1
Frame initiates connection closing	TCP	1
Time To Live	IPv4	1
Connection finish (FIN)	TCP	2
Connection establishment acknowledge (SYN+ACK)	TCP	1
Connection establishment request (SYN)	TCP	1

Interpretation

The TCP entries are consistent with connection establishment and termination.

The important point is that the Expert Information window did not display packet-level errors or warnings in the reviewed results.

Screenshot Evidence



19. Security Findings

The analysis produced several security-relevant observations.

Finding 1 – Unencrypted HTTP

The target uses HTTP rather than HTTPS.

Risk:

HTTP communication can be observed in plaintext by someone able to capture the traffic.

Recommendation:

Use HTTPS with properly configured TLS.

Finding 2 – Server Version Disclosure

The HTTP response exposed:

Apache/2.2.8 (Ubuntu)

and:

PHP/5.2.4-2ubuntu5.10

Risk:

Exposing software versions can provide useful information for attackers performing reconnaissance.

Recommendation:

Keep server software updated and minimize unnecessary version disclosure.

Finding 3 – DNS Server Failure

The DNS query for:

metasploitable.localdomain

received a:

Server failure

response.

Risk:

Repeated DNS failures can indicate configuration problems or service availability issues.

Recommendation:

Review DNS configuration, records and service health.

Finding 4 – Multiple Network Services

The earlier Nmap assessment of the same Metasploitable machine showed numerous open services.

This provides useful context for the packet analysis because the system is intentionally configured as a vulnerable laboratory target.

Recommendation:

In a production environment, disable unnecessary services and restrict access using firewall rules and network segmentation.

20. Mitigation Recommendations

The following security improvements are recommended:

1. Use HTTPS instead of HTTP.
2. Keep Apache and PHP updated to supported versions.
3. Avoid unnecessary disclosure of server software versions.
4. Review DNS configuration and investigate repeated Server Failure responses.
5. Disable unnecessary network services.

6. Restrict exposed services using firewall rules.
 7. Segment sensitive systems from untrusted networks.
 8. Monitor DNS and HTTP traffic for unusual patterns.
 9. Use packet-analysis tools during incident investigations.
 10. Maintain regular vulnerability assessments and security monitoring.
-

21. Overall Analysis

The Wireshark exercise successfully demonstrated how packet analysis can be used to understand network communication.

The capture allowed identification of:

ARP

DNS

TCP

HTTP

ICMP

NBNS

SSDP

DHCP

The most important traffic for this task was:

DNS

TCP

HTTP

The TCP handshake demonstrated how reliable connections are established.

The HTTP analysis demonstrated how application-layer information can be exposed through unencrypted communication.

The DNS analysis demonstrated how hostname resolution works and also revealed a Server Failure response.

The Expert Information feature provided additional information about TCP connection establishment and termination.

22. Conclusion

This task provided practical experience in network packet capture and analysis using Wireshark.

The analysis successfully demonstrated:

- Capturing packets from a network interface
- Identifying network protocols
- Analyzing DNS requests and responses
- Understanding HTTP communication
- Examining TCP packet structure
- Understanding SYN, ACK and FIN flags
- Observing the TCP three-way handshake
- Inspecting packet layers
- Identifying an anomalous DNS response
- Using Wireshark Expert Information
- Evaluating security implications of observed network traffic

The analysis did not identify a confirmed malformed packet or packet-level error. Instead, the DNS Server Failure response was documented as an anomalous event based directly on the captured evidence.
